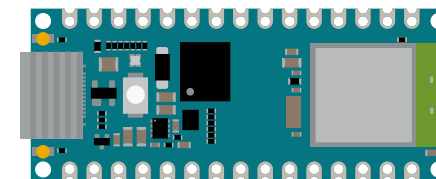


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Nano ESP32

The Arduino Nano ESP32 is the first ever Arduino board based on a ESP32 microcontroller from [Espressif](#), the **NORA-W106 module** from u-blox®. USB-C® connector, 16 MB (128 Mbit) of Flash, support for MicroPython & Arduino Cloud enabled, it is a very versatile development board.

This board is a perfect entry point to learn MicroPython, dive into it with our free course: [MicroPython 101](#)

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Here you will find the technical specifications for the Arduino® Nano ESP32.

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Please read: operating voltage

The microcontroller on the **Arduino Nano ESP32** runs at 3.3 V, which means that you

Nano 33 BLE Sense Rev2

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exceeded. Connecting higher voltage signals, like the 5 V commonly used with the other Arduino boards, will damage the Nano ESP32.

Note that this board does not have a 5V pin, instead, it has a VBUS pin. The VBUS pin provides 5 V as long as it is powered via USB. Powering via VIN will not enable the VBUS pin, meaning you have no option to receive 5 V unless powered with USB. For communication with 5 V devices, you can opt to use a logic level translator.

Board	
Name	Arduino® Nano ESP32
SKU	ABX00083 / ABX00092 / ABX00083_CN / ABX00092_CN
Microcontroller	
u-blox® NORA-W106-10B (ESP32-S3)	
USB connector	
USB-C®	
Pins	

Digital I/O Pins	14
Analog input pins	8
PWM pins	All pins (max 5 simultaneously)
External interrupts	All digital pins
Connectivity	
Wi-Fi®	Wi-Fi 4 (802.11b/g/n) 2.4 GHz
Bluetooth®	Bluetooth 5.0 (BLE support)
Communication	
UART	2x
I2C	1x, A4 (SDA), A5 (SCL)
SPI	D11 (COPI), D12 (CIPO), D13 (SCK). Use any GPIO for Chip Select (CS).
Power	
I/O Voltage	3.3 V
Input voltage	5-21 V

Source Current per I/O Pin	40 mA
Sink Current per I/O Pin	28 mA
Clock speed	
Processor	up to 240 MHz
Memory	
ROM	384 kB
SRAM	512 kB
PSRAM	8 MB
External Flash	128 Mbit (16 MB)
Dimensions	
Width	18 mm
Length	45 mm

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